Problem 18.23

The accumulated value just after the last payment under a 12-year annuity-immediate of 1000 per year, paying interest at the rate of 5% per annum effective, is to be used to purchase a perpetuity at an interest rate of 6%, with first payment to be made 1 year after the last payment under the annuity. Determine the size of the payments under the perpetuity.

Solution

First find the accumulated value of the 12-year annuity-immediate at 5%. Since payments of 1000 are made at the end of each year for 12 years,

$$AV = 1000 s_{\overline{12}|0.05}.$$

Using

$$s_{\overline{12}|0.05} = \frac{(1.05)^{12} - 1}{0.05},$$

we have

$$AV = 1000\left(\frac{(1.05)^{12} - 1}{0.05}\right).$$

This accumulated value is used to purchase a perpetuity-immediate at 6%. If the perpetuity payment is K, then its value one year before the first perpetuity payment is

$$K a_{\overline{\infty}|0.06} = K\left(\frac{1}{0.06}\right).$$

Thus,

$$1000\left(\frac{(1.05)^{12} - 1}{0.05}\right) = K\left(\frac{1}{0.06}\right).$$

Solving for K,

$$K = 1000\left(\frac{(1.05)^{12} - 1}{0.05}\right)(0.06).$$

Therefore,

$$K = 955.0275912.$$

Thus, the perpetuity payments are

$$\boxed{955.03}.$$